CS 301 · MEETING 12 · A DEEPER LOOK 1

Memory and Computation

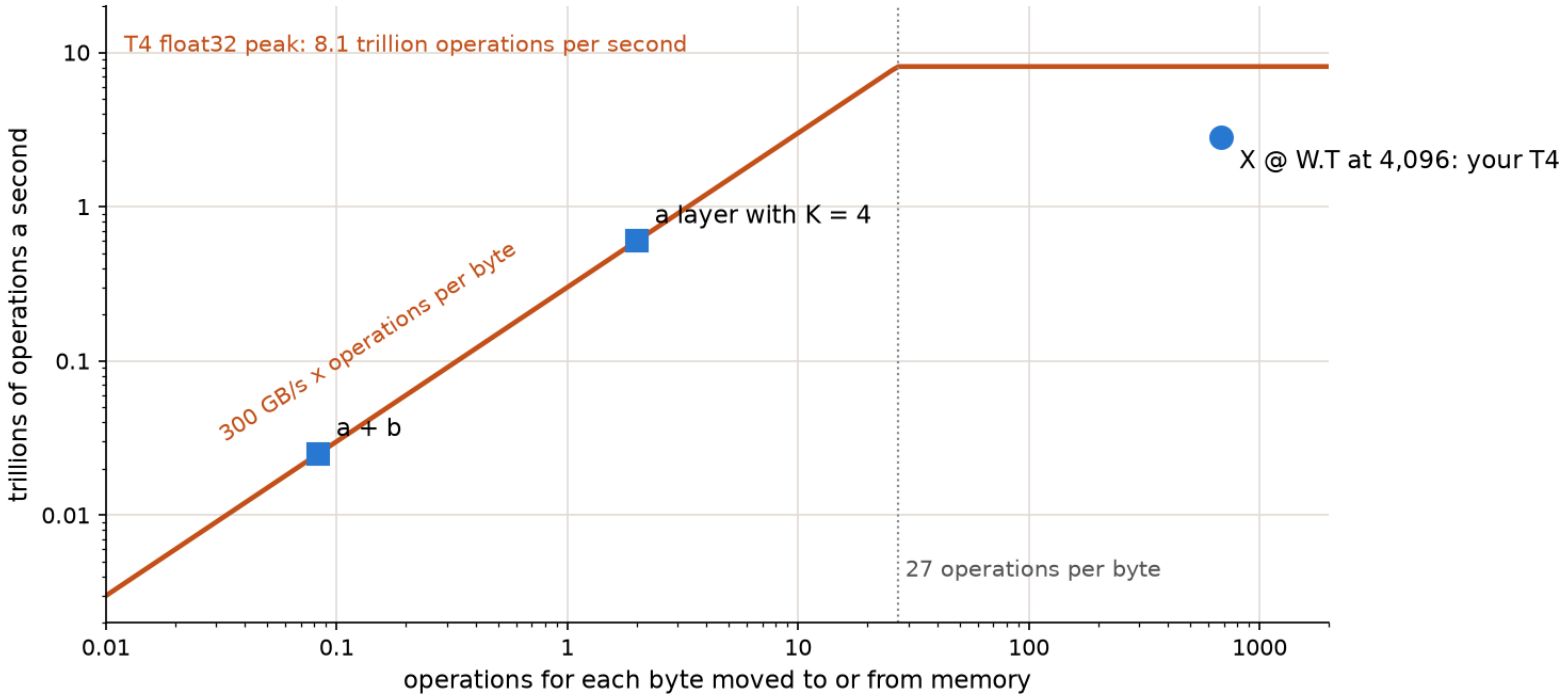
Operation counts and memory traffic: the roofline, blocking, and GPU transfers.

Operations per byte

$$a + b\text{: } \frac{1 \text{ operation}}{12 \text{ bytes}} \qquad X\,W^T \text{ at size } N\text{: } \frac{2N^3 \text{ operations}}{3 \times 4N^2 \text{ bytes}} = \frac{N}{6}$$

- Idealized traffic at N = 4,096: **683 operations/byte**
- Addition uses each input once; the product reuses it N times
- Operations per byte select the lower roofline bound

The roofline of the T4



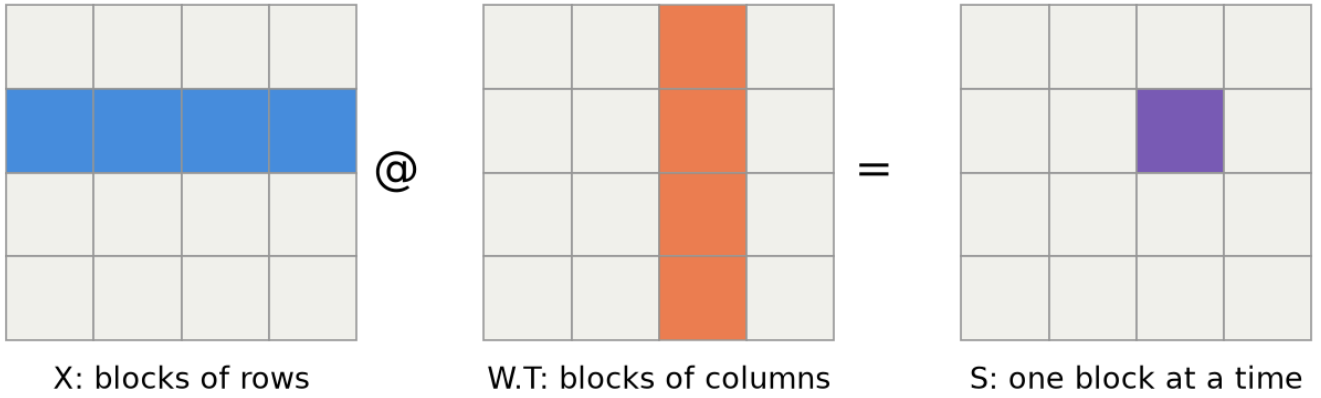
- speed ≤ min(peak rate, 300 GB/s × operations per byte)

The T4 reached a third of its peak

the T4	operations a second	the product at size 4,096
its data sheet: the peak	8.1 trillion	0.017 s
published benchmarks of cuBLAS on Colab	4.6 trillion	0.030 s
your posts, median of 12	**2.8 trillion**	**0.049 s**

- Compute-bound region; measured rate below the theoretical peak
- $2 \times 4{,}096^3$: 137 billion operations in each product

Computing the product in blocks



- The library computes S one block at a time
- Reuse cached input blocks for many multiplications
- Reordering preserves O(nKm) and reduces memory traffic

When copying takes longer than computing

computation	copies across the link	GPU compute time
X @ W.T at size 4,096 on the T4	12.6 milliseconds	48.7 milliseconds
a + b on 16.8 million numbers on the T4	12.6 milliseconds	0.671 milliseconds

- Matrix product: copying takes less time than multiplication
- a + b: copying takes far longer than adding
- Keep data on the GPU across repeated computations

Try it on the T4

```
n = 2 ** 26                                          # 67 million numbers in each array
a = torch.rand(n, device="cuda")
b = torch.rand(n, device="cuda")
c = torch.empty(n, device="cuda")
torch.add(a, b, out=c); torch.cuda.synchronize()     # once, not measured
start = time.perf_counter()
for _ in range(20):
    torch.add(a, b, out=c)
torch.cuda.synchronize()
seconds = (time.perf_counter() - start) / 20
print(f"{n / seconds / 1e9:.1f} billion additions a second, {12 * n / seconds / 1e9:.0f} GB/s")
```

- Roofline prediction: at most 300 GB/s, or 25 billion additions/second

What decides the speed

$$\text{time} \approx \max\left(\frac{\text{operations}}{\text{peak rate}}, \frac{\text{bytes moved}}{\text{bandwidth}}\right)$$

- O( ): growth of the operation count
- Operations per byte: memory-bound or compute-bound
- Data location: cache, main memory, or GPU memory